

CBSE CLASS 10 OFFLINE STUDY PACK

2026–27 | Maths • Science • Revision • Practice • Planning

Download once, study offline. This starter pack includes formula revision, key science concepts, the first 20 elements, mind-map prompts, practice questions with explanations, and printable trackers.

Independent study resource — not an official CBSE or NCERT publication.

How to use this pack

- Learn one topic, then close the notes and recall it from memory.
- Attempt the quiz before reading the answer key.
- Write every mistake in the mistake log and retry it later.
- Use the weekly planner to set realistic goals and include breaks.
- Download official CBSE/NCERT PDFs while online, then keep them on your device.

PDF limitation: this file works offline, but it cannot automatically score quizzes or save a digital dashboard. Use the score and attempt trackers below.

Official resources to download for offline use

Resource	Official page
Class X Sample Papers + Marking Schemes 2026–27	https://cbseacademic.nic.in/SQP_CLASSX_2026-27.html
Previous-year question papers	https://www.cbse.gov.in/cbsenew/question-paper.html
CBSE curriculum portal	https://cbseacademic.nic.in/curriculum_2027.html
NCERT textbooks/resources	https://ncert.nic.in/

Mathematics — quick formula sheet

Real numbers

- $\text{HCF}(a,b) \times \text{LCM}(a,b) = a \times b$ for positive integers.
- Euclid's division lemma: $a = bq + r$, where $0 \leq r < b$.
- A reduced fraction p/q has a terminating decimal if q has only 2 and/or 5 as prime factors.

Polynomials and quadratic equations

- For $ax^2 + bx + c$, sum of zeroes = $-b/a$; product = c/a .
- Quadratic formula: $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$.
- Discriminant $D = b^2 - 4ac$: positive $\rightarrow$ two distinct real roots; zero $\rightarrow$ equal real roots; negative $\rightarrow$ no real roots.

Arithmetic progressions

- n th term: $a_n = a + (n-1)d$.
- Sum: $S_n = \frac{n}{2} [2a + (n-1)d] = \frac{n}{2}(a + l)$.

Coordinate geometry

- Distance = $\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$.
- Midpoint = $\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right)$.
- Internal section $m:n$: $\left(\frac{mx_2 + nx_1}{m+n}, \frac{my_2 + ny_1}{m+n}\right)$.

Triangles and trigonometry

- Pythagoras: $\text{hypotenuse}^2 = \text{base}^2 + \text{perpendicular}^2$.
- $\sin \theta = \text{perpendicular/hypotenuse}$; $\cos \theta = \text{base/hypotenuse}$; $\tan \theta = \text{perpendicular/base}$.
- $\sin^2 \theta + \cos^2 \theta = 1$; $\tan \theta = \sin \theta / \cos \theta$.
- $\sin(90^\circ - \theta) = \cos \theta$; $\tan(90^\circ - \theta) = \cot \theta$.

Circles, surface areas and volumes

- Circle: circumference = $2\pi r$; area = πr^2 .
- Sector area = $\frac{\theta}{360^\circ} \pi r^2$; arc length = $\frac{\theta}{360^\circ} 2\pi r$.
- Cylinder: CSA = $2\pi rh$; TSA = $2\pi r(h+r)$; volume = $\pi r^2 h$.
- Cone: volume = $\frac{1}{3} \pi r^2 h$; CSA = πrl ; $l = \sqrt{r^2 + h^2}$.
- Sphere: area = $4\pi r^2$; volume = $\frac{4}{3} \pi r^3$. Hemisphere TSA = $3\pi r^2$; volume = $\frac{2}{3} \pi r^3$.

Statistics and probability

- Grouped mean: $\bar{x} = \frac{\sum f_i x_i}{\sum f_i}$; class mark = $(\text{lower} + \text{upper limit})/2$.
- $P(E) = \text{favourable outcomes} / \text{total equally likely outcomes}$.
- $0 \leq P(E) \leq 1$; $P(\text{not } E) = 1 - P(E)$.

Science — quick revision sheet

Physics: electricity

- Current: $I = Q/t$; potential difference: $V = W/Q$; Ohm's law: $V = IR$.
- Series resistance: $R = R_1 + R_2 + \dots$
- Parallel resistance: $1/R = 1/R_1 + 1/R_2 + \dots$
- Power: $P = VI = I^2R = V^2/R$; energy: $E = Pt$.
- $1 \text{ kWh} = 3.6 \times 10^6 \text{ J}$.

Physics: light and human eye

- Mirror formula: $1/f = 1/v + 1/u$ (Cartesian sign convention).
- Mirror magnification: $m = h_2/h_1 = -v/u$.
- Lens formula: $1/f = 1/v - 1/u$ (Cartesian sign convention).
- Lens power: $P = 1/f$, with f in metres; unit is dioptre (D).
- Myopia is corrected with a concave lens; hypermetropia with a convex lens.

Chemistry

- Balance equations using coefficients, not by changing formula subscripts.
- Oxidation commonly means gain of oxygen/loss of hydrogen; reduction commonly means loss of oxygen/gain of hydrogen.
- At about 25°C , pH below 7 is acidic, 7 neutral, and above 7 basic.
- A more reactive metal can displace a less reactive metal from its salt solution.
- Ionic compounds usually conduct electricity when molten or dissolved, but not as solids.

Biology

- Photosynthesis (simplified): $\text{carbon dioxide} + \text{water} \rightarrow \text{glucose} + \text{oxygen}$, with light and chlorophyll.
- Xylem transports water/minerals; phloem transports food.
- Aerobic respiration uses oxygen; anaerobic respiration occurs without oxygen.
- Neurons carry nerve impulses; hormones act as chemical messengers.
- A reflex arc coordinates a rapid response.

Periodic table — first 20 elements

Atomic no.	Symbol	Element
1	H	Hydrogen
2	He	Helium
3	Li	Lithium
4	Be	Beryllium
5	B	Boron
6	C	Carbon
7	N	Nitrogen
8	O	Oxygen
9	F	Fluorine
10	Ne	Neon

Atomic no.	Symbol	Element
11	Na	Sodium
12	Mg	Magnesium
13	Al	Aluminium
14	Si	Silicon
15	P	Phosphorus
16	S	Sulfur
17	Cl	Chlorine
18	Ar	Argon
19	K	Potassium
20	Ca	Calcium

Mind-map starters

Maths

- Quadratic equations: standard form → methods → discriminant → nature of roots → word problems.
- Trigonometry: ratios → standard values → identities → complementary angles → heights and distances.
- Statistics: class intervals → class marks → cumulative frequency → mean/median/mode.
- Surface areas and volumes: identify solid → list dimensions → select formula → substitute → units.

Science

- Life processes: nutrition → respiration → transportation → excretion.
- Electricity: current → potential difference → resistance → circuits → power/energy.
- Carbon compounds: covalent bonds → homologous series → functional groups → reactions → ethanol/ethanoic acid → soaps.
- Control and coordination: nervous system → neuron → reflex arc → brain/spinal cord → hormones → plant responses.

Practice quiz — try first

1. HCF = 6 and LCM = 180. Product of the two integers?

A) 30 B) 186 C) 1080 D) 900

2. For $x^2 - 5x + 6 = 0$, sum of roots?

A) -5 B) 5 C) 6 D) -6

3. AP with $a = 4$, $d = 3$. nth term?

A) $3n+1$ B) $4n+3$ C) $3n+4$ D) $n+7$

4. If $\sin \theta = 3/5$ for acute θ , $\cos \theta$?

A) $3/4$ B) $4/5$ C) $5/4$ D) $5/3$

5. A lens has focal length +0.5 m. Power?

A) +2 D B) -2 D C) +0.5 D D) -0.5 D

6. Plant tissue that transports water and minerals?

A) Phloem B) Xylem C) Epidermis D) Stomata

7. A solution with pH 3 is:

A) Acidic B) Neutral C) Basic D) Always salt

8. Gas commonly released during photosynthesis?

A) Nitrogen B) Oxygen C) Hydrogen D) Methane

9. Electric power formula?

A) $P=VI$ B) $P=V/I$ C) $P=I/V$ D) $P=Q/t$ only

10. Probability of an impossible event?

A) 1 B) $1/2$ C) 0 D) -1

Answer key and explanations

1. **Answer: C** — Product = HCF $\times$ LCM = 1080.
2. **Answer: B** — Sum = $-b/a = 5$.
3. **Answer: A** — $a_n = a + (n-1)d = 3n+1$.
4. **Answer: B** — Using the 3–4–5 triangle, $\cos \theta = 4/5$.
5. **Answer: A** — $P = 1/f = +2 \text{ D}$.
6. **Answer: B** — Xylem transports water and minerals.
7. **Answer: A** — pH below 7 is acidic.
8. **Answer: B** — Oxygen is released during photosynthesis.
9. **Answer: A** — $P = VI$; also $P = I^2R = V^2/R$.
10. **Answer: C** — An impossible event has probability 0.

Score: ____ / 10 Accuracy: ____ % Date: _____

Questions to retry: _____

Mistake log

Date	Chapter / question	Why was it wrong?	Correct method / next step

Weekly study planner

Set achievable tasks. Include schoolwork, breaks and sleep; adjust this to your timetable.

Day	Maths goal	Science goal	Practice / revision	Done
Monday				■
Tuesday				■
Wednesday				■
Thursday				■
Friday				■
Saturday				■
Sunday				■

Attempt tracker

Attempt	Date	Test / chapter	Marks	One thing to improve
1				
2				
3				
4				
5				
6				
7				

Board-exam practice routine

- Read each question twice and underline command words: define, explain, calculate, compare, justify.
- For numericals: Given → Formula → Substitution → Calculation → Unit.
- For science answers, use clear points and the key terms from NCERT.
- For diagrams, label neatly and include a title when useful.
- If stuck, move on and return later instead of losing time on one question.
- After a mock test, classify mistakes: concept, calculation, reading, memory, or time management.

After Class 10 / 12 — counselling checklist

- After Class 10, compare Science, Commerce, Humanities and vocational options using interests, subjects, costs and local availability.
- Ask which subjects you will study daily, what support is available, and what the fees/commute will be.
- After Class 12, compare course eligibility, entrance exams, total costs, location and official deadlines.
- Discuss choices with a trusted teacher, guardian or school counsellor. Verify details on official admission portals; avoid guaranteed-selection claims.

Offline readiness checklist

- Download this PDF and open it once to confirm it works.
- While online, download official sample papers, marking schemes, syllabus, NCERT books and previous-year papers.
- Keep the downloaded files in a named folder on your phone or computer.
- Saved PDFs work offline; links inside them require internet to open external pages.
- This pack is a starter revision aid, not a complete replacement for the current syllabus or textbooks.